

Problem Set 4

Applied Stats II

Due: April 10, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in **.pdf** form.
- This problem set is due before 23:59 on Friday April 10, 2026. No late assignments will be accepted.

Question 1

We're interested in modeling the historical causes of child mortality. We have data from 26855 children born in Skellefteå, Sweden from 1850 to 1884. Using the "child" dataset in the **eha** library, fit a Cox Proportional Hazard model using mother's age and infant's gender as covariates. Present and interpret the output.

Question 2

We want to estimate how the amount of damage caused by a disaster relates to (1) whether disaster relief is provided and (2) the amount of relief that is provided conditional on a donation occurring when a disaster hits a given country. Estimate a Heckman selection model using the **disasters_response** data on GitHub in which **binContribution** is the outcome for the selection equation, and **originalContributionMillionUSDLogged** is the outcome for the second equation. Use **occurrences + deathsEM + normalizedDamageEMLogged** as your input variables for both equations. Present and interpret the output.

Question 1: Answer

Load the data on child mortality by mother's background and child gender

```
1 data("child")
```

Fit a Cox Proportional Hazard model and summarize the results

```
1 child_surv <- with(child, Surv(enter, exit, event))
2 cox_mortality <- coxph(child_surv ~ sex + m.age, data = child)
3 summary(cox_mortality)
```

Call:

```
coxph(formula = child_surv ~ sex + m.age, data = child)
```

n= 26574, number of events= 5616

	coef	exp(coef)	se(coef)	z	Pr(> z)	
sexfemale	-0.082215	0.921074	0.026743	-3.074	0.002110	**
m.age	0.007617	1.007646	0.002128	3.580	0.000344	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
sexfemale	0.9211	1.0857	0.874	0.9706
m.age	1.0076	0.9924	1.003	1.0119

Concordance= 0.519 (se = 0.004)

Likelihood ratio test= 22.52 on 2 df, p=1e-05

Wald test = 22.52 on 2 df, p=1e-05

Score (logrank) test = 22.53 on 2 df, p=1e-05

There is a 0.082215 decrease in the expected log of the hazard for female babies compared to male, holding mother's age constant. The hazard ratio of female babies is 0.92 to that of male babies, i.e. female babies are less likely to die (92 female babies die for every 100 male babies/female deaths are 8 percent lower).

For every one unit increase in the mother's age, there is a 0.007617 increase in the expected log of the hazard for babies, holding sex constant. For every one unit increase in the mother's age, there is a 0.76 percent increase in the expected hazard for babies, holding sex constant.

Question 2: Answer

Load the disaster data

```
1 disaster_data <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsII_2026/refs/heads/main/datasets/disaster_response.csv")
```

The OLS model

```
1 base_heck <- lm(binContribution ~ occurrences + deathsEM +  
  normalizedDamageEMLogged, data = disaster_data)  
2 base_heck
```

Call:

```
lm(formula = binContribution ~ occurrences + deathsEM + normalizedDamageEMLogged,  
    data = disaster_data)
```

Coefficients:

(Intercept)	occurrences	deathsEM
0.092691	0.003520	0.003854
normalizedDamageEMLogged		
	0.002549	

Interpretation:

A one unit increase in occurrences is associated with a 0.3520 percentage point increase in the probability of donation, on average and holding all else constant.

A one unit increase in deathsEM is associated with a 0.3854 percentage point increase in the probability of donation, on average and holding all else constant.

A one unit increase in normalizedDamageEMLogged is associated with a 0.2549 percentage point increase in the probability of donation, on average and holding all else constant.

```
1 heckman_model <- heckit(selection = binContribution ~ occurrences + deathsEM  
  + normalizedDamageEMLogged,  
2                        outcome = originalContributionMillionUSDLogged ~  
  occurrences + deathsEM + normalizedDamageEMLogged,  
3                        data = disaster_data)  
4  
5 summary(heckman_model)
```

Heckman Selection Model:

```
-----
Tobit 2 model (sample selection model)
2-step Heckman / heckit estimation
49096 observations (45848 censored and 3248 observed)
11 free parameters (df = 49086)
Probit selection equation:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)    -1.3019633   0.0180138  -72.28   <2e-16 ***
occurrences      0.0191534   0.0016550   11.57   <2e-16 ***
deathsEM         0.0114710   0.0007178    15.98   <2e-16 ***
normalizedDamageEMLogged 0.0211062   0.0009035    23.36   <2e-16 ***
Outcome equation:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)      9.76625    6.04404    1.616   0.1061
occurrences     -0.08634    0.05492   -1.572   0.1159
deathsEM        -0.03298    0.02569   -1.284   0.1993
normalizedDamageEMLogged -0.11420    0.06231   -1.833   0.0668 .
Multiple R-Squared:0.0546, Adjusted R-Squared:0.0534
Error terms:
      Estimate Std. Error t value Pr(>|t|)
invMillsRatio   -6.313      3.420   -1.846   0.0649 .
sigma           5.992         NA      NA      NA
rho            -1.053         NA      NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
-----
```

None of the covariates are statistically different than zero in outcome equation.

The p-value of invMillsRatio is 0.0649, which is above our critical value of 0.05. Therefore, we fail to reject the null hypothesis that there is no selection bias happening in OLS that needs to be accounted for with a Heckman selection model. Therefore, we can continue to use the OLS model.

This indicates that while there is a limited selection of instances in which donation occurred, it is not biasing the results in predicting the amount of the donation.